UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

STATISTICAL SIGNAL PROCESSING - *Prova scritta del 13/02/2025*

EXERCISE

We observe N samples of a realization of the discrete-time random process $X[n] = A\cos(2\pi F_0 n) + B\sin(2\pi F_0 n) + W[n]$, for n=0, 1,…, N-1, where N is an **even number**, and $F_0 = 1/4$ is the known digital frequency. A and B are unknown deterministic parameters. $\{W[n]; n = 0,1,\cdots,N-1\}$ are independent and identically distributed (IID) random variables modeling additive noise; each noise sample is Gaussian distributed, with zero mean and known variance σ_W^2.

(1) Derive the joint probability density function (pdf) $f_{\mathbf{X}}(\mathbf{x}; A, B)$ and the log-likelihood function $LL(A,B) = \ln f_{\mathbf{X}}(\mathbf{x}; A, B)$ of the observed vector $\mathbf{X} = [X[0]\ X[1] \cdots X[N-1]]^T$.

(2) Derive the joint maximum likelihood (ML) estimator of the unknown parameters A and B.

(3) Derive the bias and the mean square error (MSE) of the ML estimators of A and B and determine whether the estimators are unbiased and consistent.

(4) Assume now that also the noise power σ_W^2 is unknown: derive the joint ML estimators of A, B and σ_W^2 and calculate the bias of the ML estimate of σ_W^2.

Hint: If $\mathbf{P}$ is a projection matrix of rank K (note that the rank K is equal to the dimension of the projection subspace) and $\mathbf{X}$ is AWGN, then $E\{\mathbf{X}^T\mathbf{P}\mathbf{X}\} = K\sigma_W^2$.

Solution:

(1) *Derive the joint probability density function (pdf)* $f_{\mathbf{X}}(\mathbf{x}; A, B)$ *and the log-likelihood function* $LL(A,B) = \ln f_{\mathbf{X}}(\mathbf{x}; A, B)$ *of the observed vector* $\mathbf{X} = [X[0]\ X[1] \cdots X[N-1]]^T$.

$$X[n] = A\cos\left(2\pi F_0 n\right) + B\sin\left(2\pi F_0 n\right) + W[n] = A\cos\left(\frac{\pi}{2}n\right) + B\sin\left(\frac{\pi}{2}n\right) + W[n], \quad n = 0,1,\cdots,N-1$$

$$\mathbf{X} = A\mathbf{c} + B\mathbf{s} + \mathbf{W} \in N\left(A\mathbf{c} + B\mathbf{s}, \sigma_W^2 \mathbf{I}\right)$$

where:

$$\mathbf{c} \triangleq \begin{bmatrix} \cos(0) \\ \cos\left(\frac{\pi}{2}\right) \\ \vdots \\ \cos\left(\frac{\pi}{2}(N-1)\right) \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad \mathbf{s} \triangleq \begin{bmatrix} \sin(0) \\ \sin\left(\frac{\pi}{2}\right) \\ \vdots \\ \sin\left(\frac{\pi}{2}(N-1)\right) \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ -1 \\ 0 \\ 1 \\ \vdots \\ -1 \end{bmatrix}, \quad \Rightarrow \quad \|\mathbf{c}\|^2 = N/2, \quad \|\mathbf{s}\|^2 = N/2, \quad \mathbf{c}^T\mathbf{s} = 0$$

The data vector $\mathbf{X}$ can be expressed as:

$$\mathbf{X} = \begin{bmatrix} \mathbf{c} & \mathbf{s} \end{bmatrix} \begin{bmatrix} A \\ B \end{bmatrix} + \mathbf{W} = \mathbf{H}\boldsymbol{\theta} + \mathbf{W}, \quad \text{where } \underset{N\times 2}{\mathbf{H}} \triangleq \begin{bmatrix} \mathbf{c} & \mathbf{s} \end{bmatrix} \text{ and } \underset{2\times 1}{\boldsymbol{\theta}} \triangleq \begin{bmatrix} A \\ B \end{bmatrix}$$

The pdf and the log-LF are given by:

$$\mathbf{X} \in N\left(\mathbf{H}\boldsymbol{\theta}, \sigma_W^2 \mathbf{I}\right)$$

$$LL(A,B) = \ln f_{\mathbf{X}}(\mathbf{x}; A, B) = k - \frac{1}{2\sigma_W^2}\|\mathbf{X} - \mathbf{H}\boldsymbol{\theta}\|^2 = k - \frac{1}{2\sigma_W^2}\sum_{n=0}^{N-1}\left(X[n] - A\cos\left(\frac{\pi}{2}n\right) - B\sin\left(\frac{\pi}{2}n\right)\right)^2$$

where k is an irrelevant constant (only when σ_W^2 is known).

(2) *Derive the joint maximum likelihood (ML) estimator of the unknown parameters A and B.*

Since the noise is AWGN, the ML estimator coincide with the LS estimator:

$$\hat{\boldsymbol{\theta}}_{ML} = \begin{bmatrix} \hat{A}_{ML} \\ \hat{B}_{ML} \end{bmatrix} = \hat{\boldsymbol{\theta}}_{LS} = \mathbf{H}^{\#}\mathbf{X} = \left(\mathbf{H}^T\mathbf{H}\right)^{-1}\mathbf{H}^T\mathbf{X}$$

$$\mathbf{H}^T\mathbf{H} = \begin{bmatrix} \mathbf{c}^T \\ \mathbf{s}^T \end{bmatrix}\begin{bmatrix} \mathbf{c} & \mathbf{s} \end{bmatrix} = \begin{bmatrix} \mathbf{c}^T\mathbf{c} & \mathbf{c}^T\mathbf{s} \\ \mathbf{s}^T\mathbf{c} & \mathbf{s}^T\mathbf{s} \end{bmatrix} = \begin{bmatrix} N/2 & 0 \\ 0 & N/2 \end{bmatrix} \Rightarrow \left(\mathbf{H}^T\mathbf{H}\right)^{-1} = \begin{bmatrix} N/2 & 0 \\ 0 & N/2 \end{bmatrix}^{-1} = \begin{bmatrix} 2/N & 0 \\ 0 & 2/N \end{bmatrix} = \frac{2}{N}\mathbf{I}$$

$$\hat{\boldsymbol{\theta}}_{ML} = \begin{bmatrix} \hat{A}_{ML} \\ \hat{B}_{ML} \end{bmatrix} = \hat{\boldsymbol{\theta}}_{LS} = \frac{1}{N}\mathbf{I}\begin{bmatrix} \mathbf{c}^T \\ \mathbf{s}^T \end{bmatrix}\mathbf{X} = \frac{2}{N}\begin{bmatrix} \mathbf{c}^T\mathbf{X} \\ \mathbf{s}^T\mathbf{X} \end{bmatrix}$$

$$\hat{A}_{ML} = \frac{2}{N}\mathbf{c}^T\mathbf{X} = \frac{2}{N}\sum_{n=0}^{N-1} X[n]\cos\left(\frac{\pi}{2}n\right), \qquad \hat{B}_{ML} = \frac{2}{N}\mathbf{s}^T\mathbf{X} = \frac{2}{N}\sum_{n=0}^{N-1} X[n]\sin\left(\frac{\pi}{2}n\right)$$

We could have easily derived the same results by calculating the derivative of $LL(A,B)$ with respect to A and B and solving the two ML equations, that are decoupled.

(3) *Derive the bias and the mean square error (MSE) of the ML estimators of A and B and determine whether the estimators are unbiased and consistent.*

The MSE of the two estimators can be derived in different ways. One is the following.

$$\hat{A}_{ML} = \frac{2}{N}\mathbf{c}^T\mathbf{X} = \frac{2}{N}\mathbf{c}^T\left(A\mathbf{c} + B\mathbf{s} + \mathbf{W}\right) = \frac{2A}{N}\mathbf{c}^T\mathbf{c} + \frac{2B}{N}\mathbf{c}^T\mathbf{s} + \frac{2}{N}\mathbf{c}^T\mathbf{W} = A + \frac{2}{N}\mathbf{c}^T\mathbf{W}$$

$$\varepsilon_A = A - \hat{A}_{ML} = -\frac{2}{N}\mathbf{c}^T\mathbf{W} \Rightarrow b\left\{\hat{A}_{ML}\right\} = E\left\{\varepsilon_A\right\} = E\left\{\left(-\frac{2}{N}\mathbf{c}^T\mathbf{W}\right)\right\} = -\frac{2}{N}\mathbf{c}^T E\left\{\mathbf{W}\right\} = 0 \quad (\textit{unbiased})$$

$$MSE\left\{\hat{A}_{ML}\right\} = E\left\{\varepsilon_A^2\right\} = E\left\{\left(-\frac{2}{N}\mathbf{c}^T\mathbf{W}\right)\left(-\frac{2}{N}\mathbf{c}^T\mathbf{W}\right)^T\right\} = \frac{4}{N^2}\mathbf{c}^T E\left\{\mathbf{W}\mathbf{W}^T\right\}\mathbf{c} = \frac{2\sigma_W^2}{N} \quad (\textit{consistent})$$

We easily derive exactly the same results for the ML estimator of *B*.

(4) *Assume now that also the noise power* σ_W^2 *is unknown: derive the joint ML estimators of A, B and* σ_W^2 *and calculate the bias of the ML estimate of* σ_W^2.

The pdf of **X** and the log-LR are given by:

$$f_{\mathbf{X}}(\mathbf{x};A,B,\sigma_W^2)=\prod_{n=0}^{N-1}\frac{1}{\sqrt{2\pi\sigma_W^2}}e^{-\frac{1}{2\sigma_W^2}\left(X[n]-A\cos\left(\frac{\pi}{2}n\right)-B\sin\left(\frac{\pi}{2}n\right)\right)^2}=\left(2\pi\right)^{-\frac{N}{2}}+\left(\sigma_W^2\right)^{-\frac{N}{2}}+e^{-\frac{1}{2\sigma_W^2}\sum_{n=0}^{N-1}\left(X[n]-A\cos\left(\frac{\pi}{2}n\right)-B\sin\left(\frac{\pi}{2}n\right)\right)^2}$$

$$\begin{aligned}LL(A,B,\sigma_W^2)&=\ln f_{\mathbf{X}}(\mathbf{x};A,B,\sigma_W^2)=-\frac{N}{2}\ln\left(2\pi\right)-\frac{N}{2}\ln\left(\sigma_W^2\right)-\frac{1}{2\sigma_W^2}\sum_{n=0}^{N-1}\left(X[n]-A\cos\left(\frac{\pi}{2}n\right)-B\sin\left(\frac{\pi}{2}n\right)\right)^2\\&=-\frac{N}{2}\ln\left(2\pi\right)-\frac{N}{2}\ln\left(\sigma_W^2\right)-\frac{1}{2\sigma_W^2}\left\|\mathbf{X}-\mathbf{H}\boldsymbol{\theta}\right\|^2\end{aligned}$$

We have already solved the two equations that we obtain from the derivative of $LL(A,B,\sigma_W^2)$ w.r.t. *A* nd *B*, now we have to derive $LL(A,B,\sigma_W^2)$ w.r.t. the noise power and solve the equation that we obtain by setting it equal to 0:

$$\frac{dLLF(\hat{A}_{ML},\hat{B}_{ML},\sigma_W^2)}{d\sigma_W^2}=\frac{d\ln f_X(\mathbf{x};\hat{A}_{ML},\hat{B}_{ML},\sigma_W^2)}{d\sigma_W^2}=-\frac{N}{2\sigma_W^2}+\frac{1}{2\sigma_W^4}\left\|\mathbf{x}-\mathbf{H}\hat{\boldsymbol{\theta}}_{ML}\right\|^2=0$$

$$\hat{\sigma}_{W,ML}^2=\frac{1}{N}\left\|\mathbf{x}-\mathbf{H}\hat{\boldsymbol{\theta}}_{ML}\right\|^2=\frac{1}{N}\left\|\mathbf{x}-\mathbf{H}\mathbf{H}^{\#}\mathbf{X}\right\|^2=\frac{1}{N}\left\|(\mathbf{I}-\mathbf{H}\mathbf{H}^{\#})\mathbf{X}\right\|^2=\frac{1}{N}\left\|(\mathbf{I}-\mathbf{P}_H)\mathbf{X}\right\|^2=\frac{1}{N}\mathbf{X}^T(\mathbf{I}-\mathbf{P}_H)\mathbf{X}$$

$$E\left\{\hat{\sigma}_{W,ML}^2\right\}=\frac{1}{N}E\left\{\mathbf{X}^T(\mathbf{I}-\mathbf{P}_H)\mathbf{X}\right\}=\frac{1}{N}(N-2)\sigma_W^2=\left(1-\frac{2}{N}\right)\sigma_W^2\quad\text{(biased, asymptotically unbiased)}$$

The ML estimator projects the received data vector **X** onto the subspace orthogonal to the signal subspace and then calculates the energy of the data after this projection, i.e. the sum of the powers of the *N*-2 components where only the noise is present. Since the signal subspace is 2D, the orthogonal noise subspace is (*N*-2)-dimensional, hence $E\left\{\mathbf{X}^T(\mathbf{I}-\mathbf{P}_H)\mathbf{X}\right\}$ is equal to $(N-2)\sigma_W^2$.